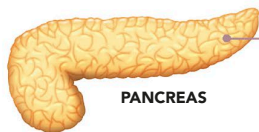


PANCREAS

The pancreas is a dual-purpose gland. It produces digestive enzymes in cells called acini, but also has an endocrine function. Within the acinar tissues are cell clusters known as islets of Langerhans, which produce hormones involved in controlling glucose (blood sugar), the body's main energy source. Beta cells make the hormone insulin, which promotes glucose uptake by body cells and speeds conversion of glucose into glycogen for storage in the liver. In this way, insulin lowers blood glucose levels.

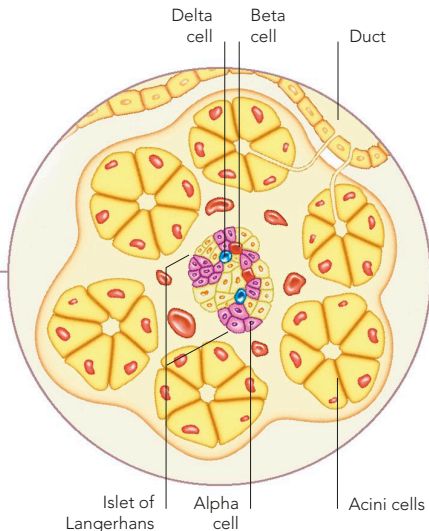


PANCREAS

PANCREATIC ISLETS

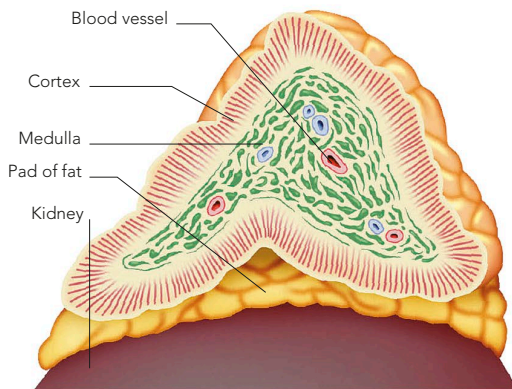
Surrounded by enzyme-producing acini cells, the tiny pancreatic islets contain three types of cells: alpha, beta, and delta. The secretions of the latter help regulate insulin and glucagon production.

Another hormone, glucagon, is produced by alpha cells and has opposing actions, raising blood glucose levels. Delta cells make somatostatin, which regulates the alpha and beta cells.



ADRENAL GLANDS

The inner layer (medulla) and outer layer (cortex) of the adrenal gland each secrete different hormones. The cortical hormones are steroids (see p.139) and include glucocorticoids, such as cortisol, which affect metabolism; mineralocorticoids, such as aldosterone, which influence salt and mineral balance; and gonadocorticoids, which act on the ovaries and testes. The medulla functions as a separate gland. Its nerve fibres link to the sympathetic nervous system, and it makes the fight-or-flight hormones, such as adrenaline.



ADRENAL ANATOMY

Each adrenal gland is shaped like a cone or pyramid and sits on top of the kidney, cushioned by a pad of fat. The glands consist of two parts: the cortex, which has three layers, and the medulla, containing nerve fibres and blood vessels.